

Objective:

Analyze sales performance and customer behavior to identify revenue drivers, customer trends, and growth opportunities.

Key Assumptions About the Data:

- Missing customer demographic details were treated as unknown values.
- LineTotal was used as the primary revenue metric.
- OnlineOrderFlag was used to distinguish online vs offline orders.
- CustomerName was derived by combining retail customer names and wholesale store names.
- Duplicate records and inactive products were excluded where necessary.
- Dates were standardized using a DimDate calendar table for time intelligence analysis.

Methodology & Derivations:

- Built a star schema model with FactSales and dimension tables (DimProduct, DimCustomer, DimTerritory, DimDate).
- Created DAX measures for Total Revenue, Total Orders, AOV, Revenue Growth %, Repeat Customers, and Retention Rate.
- Performed trend analysis by month, Country, and product category.
- Applied customer segmentation to distinguish retail and wholesale customer behavior.

Insights & Recommendations

Key Insights:

- Bikes generated the highest revenue contribution, while accessories showed strong repeat purchase behavior.
- Offline customers demonstrated higher revenue compared to online customers.
- Wholesales customers consistently outperformed Retailers, indicating stronger demand.
- Repeat customers contributed significantly more revenue than one-time buyers.

Recommendations:

1. Increase focus on customer retention programs for high-value repeat customers.
2. Promote accessory bundle offers alongside bike purchases to improve cross-selling opportunities.
3. Expand digital marketing efforts to capitalize on higher online order values.
4. Increase inventory planning for high-demand products and territories to avoid stockouts.
5. Use territory-level performance insights to guide targeted regional marketing strategies.

Conclusion:

The analysis revealed strong revenue opportunities through customer retention, Wholesalers, offline customers, and optimized product strategies. AdventureWorks can improve profitability and customer satisfaction through data-driven decision-making.